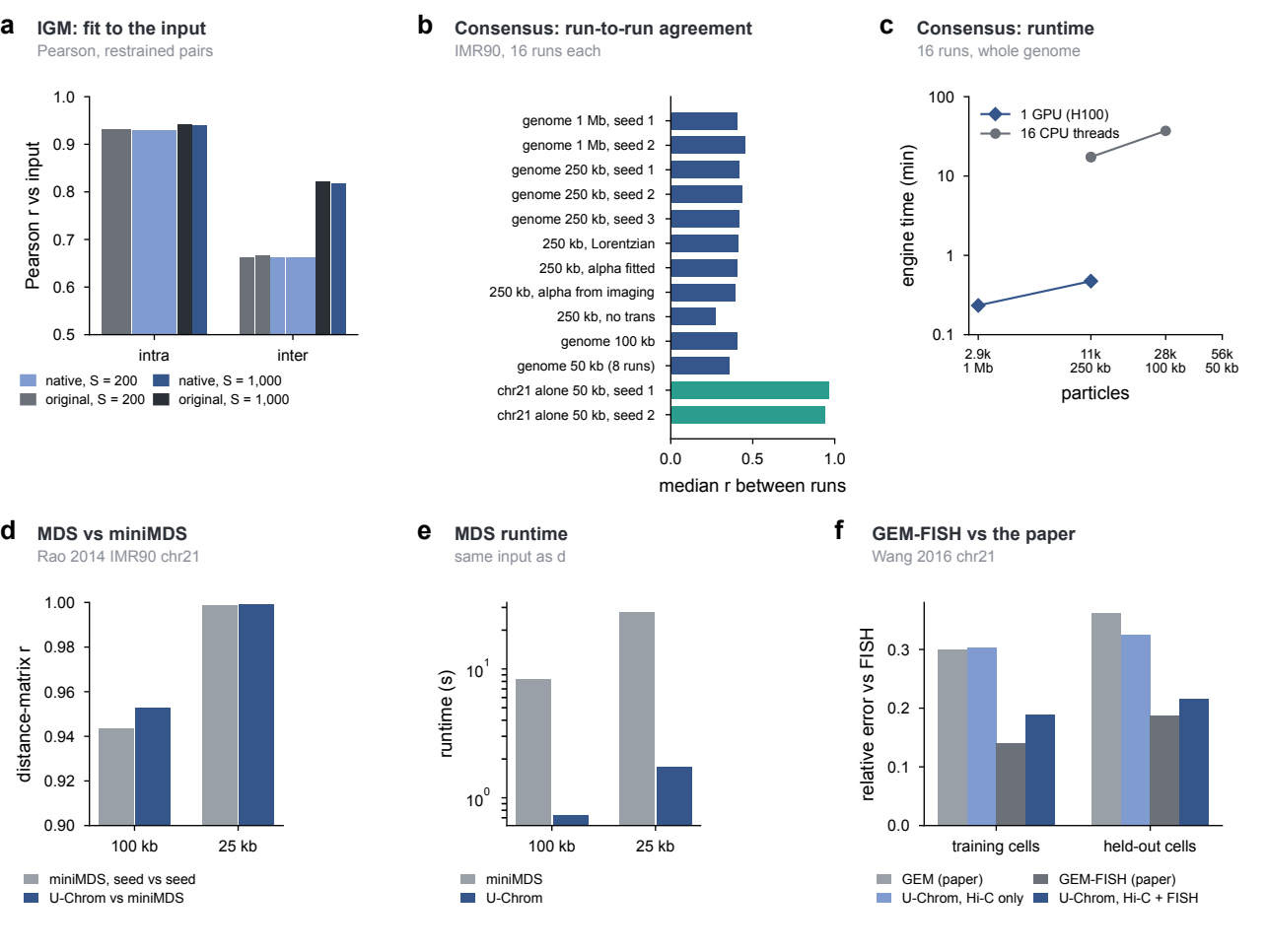
a
IGM: fit to the input
Pearson, restrained pairs
Pearson r vs input
1.0
0.9
0.8
0.7
0.6
0.5
intra
inter
native, S = 200
native, S = 1,000
original, S = 200
original, S = 1,000
b
Consensus: run-to-run agreement
IMR90, 16 runs each
genome 1 Mb, seed 1
genome 1 Mb, seed 2
genome 250 kb, seed 1
genome 250 kb, seed 2
genome 250 kb, seed 3
250 kb, Lorentzian
250 kb, alpha fitted
250 kb, alpha from imaging
250 kb, no trans
genome 100 kb
genome 50 kb (8 runs)
chr21 alone 50 kb, seed 1
chr21 alone 50 kb, seed 2
0.0
0.5
1.0
median r between runs
c
Consensus: runtime
16 runs, whole genome
1 GPU (H100)
16 CPU threads
engine time (min)
100
10
1
0.1
2.9k
1 Mb
11k
250 kb
28k
100 kb
56k
50 kb
particles
d
MDS vs miniMDS
Rao 2014 IMR90 chr21
distance-matrix r
1.00
0.98
0.96
0.94
0.92
0.90
100 kb
25 kb
miniMDS, seed vs seed
U-Chrom vs miniMDS
e
MDS runtime
same input as d
runtime (s)
10^1
10^0
100 kb
25 kb
miniMDS
U-Chrom
f
GEM-FISH vs the paper
Wang 2016 chr21
relative error vs FISH
0.3
0.2
0.1
0.0
training cells
held-out cells
GEM (paper)
U-Chrom, Hi-C only
GEM-FISH (paper)
U-Chrom, Hi-C + FISH